

Bridge Swarm IoT — three-tier architecture

EDGE — Pegasus / Isaac Sim a6o mock-simulator

UAV 1
SLAM + YOLOv8
+ planner

UAV 2
SLAM + YOLOv8
+ planner

UAV 3
SLAM + YOLOv8
+ planner

Swarm Coordinator
strip partitioning

MQTT

GATEWAY — Mosquitto MQTT v5

Mosquitto Broker
QoS 0 / 1

CLOUD — docker-compose stack

Telegraf
MQTT → Influx

InfluxDB 2.x
time series

Grafana 11
dashboards

FastAPI
/defects /report

MinIO
S3 для зображень

Operator browser ← <http://localhost:3000> / <http://localhost:8000/docs>